

Time Dependent Perturbation Theory

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Time dependent SE

$$\hat{H}|\psi(t)\rangle = i\hbar \frac{d}{dt}|\psi(t)\rangle \quad (1)$$

Time independent

$$\hat{H}|\psi\rangle = E|\psi\rangle \quad (2)$$

Start with unperturbed system

$$\hat{H}_0|\psi_k\rangle = E_k|\psi_k\rangle \quad (3)$$

Look at time evolution of a state

$$|\psi(t)\rangle = \sum_k |\psi_k^0\rangle c_k^0 e^{-i\omega_k t} \quad (4)$$

We see all of the time dependence in the phase e not the constant.
in the time-dependent eqn:

$$\hat{H}|\psi(t)\rangle = \sum_k E_k |\psi_k^0\rangle c_k^0 e^{-i\omega_k t} = i\hbar \sum_k |\psi_k^0\rangle c_k^0 - i\omega_k e^{-i\omega_k t} \quad (5)$$

Now if we act on the thing above

$$\langle \psi_l^0 | *^k E_l^0 c_l^0 e^{-i\omega_l t} = \hbar \omega_l e^{-i\omega_l t} c_l^0 \quad (6)$$

$$\omega_l = \frac{E_l}{\hbar} \quad (7)$$

We have considered a time independent Hamiltonian with all the time dependence in the perturbation.
Now we add the perturbation with a small g .

We add the perturbation $g\hat{V}(t)$

$$|\psi(t)\rangle = \sum_k |\psi_k^0\rangle e^{-i\omega_k t} (c_k^0 + g c_k^1(t) + \mathcal{O}(g^2)) \quad (8)$$

We have 'promotion' to now be a function of time.

Time-dependence

$$\left(\hat{H}_0 + g\hat{V}(t) \right) |\psi(t)\rangle = i\hbar \frac{d}{dt} |\psi(t)\rangle \quad (9)$$

At first order

$$\left(\hat{H}_0 + g\hat{V}(t) \right) \sum_k |\psi_k^0\rangle e^{-i\omega_k t} (c_k^0 + g c_k^1(t)) = i\hbar \sum_k |\psi_k^0\rangle e^{-i\omega_k t} \left(-i\omega_k (c_k^0 + g c_k^1(t)) + g \frac{dc_k^1(t)}{dt} \right) \quad (10)$$

$$g \left[\sum_k E_k |\psi_k^0\rangle c_k^1(t) e^{-i\omega_k t} + \hat{V}(t) |\psi_k^0\rangle e^{-i\omega_k t} c_k^0 \right] = g \left[i\hbar \sum_k |\psi_k^0\rangle e^{-i\omega_k t} \left(-i\omega_k c_k^1(t) + \frac{dc_k^1(t)}{dt} \right) \right] \quad (11)$$

If we act by

$$\langle \psi_l^0 | * \quad (12)$$

$$E_l e^{-i\omega_l t} c_l^1(t) + \sum_k \langle \psi_l^0 | \hat{V}(t) | \psi_k^0 \rangle e^{-i\omega_k t} c_k^0 = i\hbar e^{-i\omega_l t} \left(-i\omega_l c_l^1(t) + \frac{dc_l^1(t)}{dt} \right) \quad (13)$$

$$\frac{dc_l^1}{dt} = -\frac{i}{\hbar} \sum_k \langle \psi_l^0 | \hat{V} | \psi_k^0 \rangle c_k^0 e^{-i(\omega_k - \omega_l)t} \quad (14)$$